

TypeScript Syllabus for React JS & Next.js Developer

1.1 Introduction

- What is TypeScript
- Why TypeScript
- JavaScript vs TypeScript
- Advantages of TypeScript
- TypeScript Compilation Process
- Static Typing vs Dynamic Typing

1.2 Environment Setup

- Installing TypeScript
 - TypeScript Compiler (tsc)
 - Creating First TypeScript File
 - Running TypeScript Files
-

2. Variables

- Variable Declaration
- Type Annotation
- Type Inference

3. Primitive Types

- string
- number
- boolean
- null
- undefined

4. Arrays

- Array Types
- Array of Objects
- Readonly Arrays

5. Tuples

- Tuple Basics
- Optional Tuple Values
- Readonly Tuples

6. Objects

- Object Types
 - Nested Objects
 - Optional Properties
 - Readonly Properties
-

7. Function Basics

- Function Declaration
- Function Expression
- Arrow Functions

8. Function Parameters

- Required Parameters
- Optional Parameters
- Default Parameters
- Rest Parameters

9. Return Types

- Explicit Return Types
- Inferred Return Types

10.Function Types

- Function Signatures
- Callback Functions
- Higher Order Functions

11.Special Types

- Any Type
 - Unknown Type
 - Void Type
 - Never Type
-

12.Type Aliases

- Creating Type Aliases
 - Object Type Aliases
 - Function Type Aliases
 - Nested Types
 - Reusable Types
-

13.Interfaces

- Interface Basics
 - Optional Properties
 - Readonly Properties
 - Methods in Interfaces
 - Interface Extension
 - Multiple Interface Extension
 - Interface vs Type
-

14. Union and Intersection Types

- Union Types
 - Literal Types
 - Intersection Types
 - Combining Multiple Types
 - Discriminated Unions
-

15. Enums

- 8.1 Numeric Enums
 - 8.2 String Enums
 - 8.3 Const Enums
 - 8.4 Enum Use Cases
-

16. Type Assertions

- Type Assertions Basics
 - as Keyword
 - DOM Type Assertions
 - Non-null Assertion Operator (!)
-

17. Type Narrowing

- Type of Narrowing
- instanceof Narrowing
- in Operator
- Equality Narrowing
- Custom Type Guards
- User Defined Type Guards
- Type Predicates (value is Type)

18. Generics

- Introduction to Generics
 - Generic Functions
 - Generic Interfaces
 - Generic Type Aliases
 - Generic Constraints
 - Multiple Generics
 - Generic Utility Functions
-

19. Advanced Types

- keyof Operator
 - typeof Operator
 - Indexed Access Types
 - Lookup Types
 - Mapped Types
 - Conditional Types
 - Infer Keyword
 - Template Literal Types
-

20. Utility Types

- Partial
- Required
- Readonly
- Pick
- Omit
- Record
- Exclude

- Extract
 - NonNullable
 - ReturnType
 - Parameters
 - ConstructorParameters
 - Awaited
 - InstanceType
 - ThisParameterType
 - OmitThisParameter
-

21. Modules

- Export
 - Import
 - Named Export
 - Default Export
-

22. Classes (Basic Understanding)

- Class Basics
 - Constructor
 - Access Modifiers
 - Public
 - Private
 - protected
 - Readonly Properties
 - Implementing Interfaces
-

23. Asynchronous TypeScript

- Promise Type
 - Async Functions
 - Await Keyword
 - Typing API Responses
 - Error Handling Types
-

24. Declaration Files

- What are .d.ts Files
 - Type Definitions
 - DefinitelyTyped
 - Third-party Library Types
-